

**SAFETY DATA SHEET**

Creation Date 06-Feb-2012

Revision Date 06-Jan-2023

Revision Number 5

**1. Identification**

**Product Name** Diphosgene

**Cat No. :** AC243310000; AC243310100; AC243311000

**CAS No** 503-38-8  
**Synonyms** Trichloromethyl chloroformate

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

**Details of the supplier of the safety data sheet****Company**

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Emergency Telephone Number**

For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
**Emergency Number US:**001-201-796-7100 / **Europe:** +32 14 57 52 99  
**CHEMTREC** Tel. No.**US:**001-800-424-9300 / **Europe:**001-703-527-3887

**2. Hazard(s) identification****Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                    |              |
|------------------------------------|--------------|
| Acute oral toxicity                | Category 2   |
| Acute Inhalation Toxicity - Vapors | Category 1   |
| Skin Corrosion/Irritation          | Category 1 B |
| Serious Eye Damage/Eye Irritation  | Category 1   |

**Label Elements****Signal Word**

Danger

**Hazard Statements**

Causes severe skin burns and eye damage  
Fatal if swallowed or if inhaled

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Do not get in eyes, on skin, or on clothing  
Wear protective gloves/protective clothing/eye protection/face protection  
Do not breathe dust/fume/gas/mist/vapors/spray  
Use only outdoors or in a well-ventilated area  
Wear respiratory protection

**Response**

Immediately call a POISON CENTER or doctor/physician

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

**Skin**

Wash contaminated clothing before reuse  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

**Ingestion**

Rinse mouth  
Do NOT induce vomiting

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed  
Keep at temperatures between 2 °C and 8 °C.

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

Lachrymator (substance which increases the flow of tears)

### 3. Composition/Information on Ingredients

| Component                                    | CAS No   | Weight % |
|----------------------------------------------|----------|----------|
| Carbonochloridic acid, trichloromethyl ester | 503-38-8 | >95      |

### 4. First-aid measures

**General Advice**

Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.

**Eye Contact**

In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

**Inhalation**

Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is

|                                            |                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | required. If not breathing, give artificial respiration.                                                                                                                                                                                                                                        |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                  |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                                                                                                                           |

## 5. Fire-fighting measures

|                                         |                                                                                       |
|-----------------------------------------|---------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                              |
| <b>Flash Point</b>                      | No information available                                                              |
| <b>Method -</b>                         | No information available                                                              |
| <b>Autoignition Temperature</b>         | No information available                                                              |
| <b>Explosion Limits</b>                 |                                                                                       |
| <b>Upper</b>                            | No data available                                                                     |
| <b>Lower</b>                            | No data available                                                                     |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                              |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                              |

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Chlorine. Thermal decomposition can lead to release of irritating gases and vapors. Phosgene. Hydrogen chloride gas.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

**Health**  
4

**Flammability**  
1

**Instability**  
1

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                  |                                                                                                                                                               |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Use personal protective equipment as required. Ensure adequate ventilation. |
| <b>Environmental Precautions</b> | Should not be released into the environment. See Section 12 for additional Ecological Information.                                                            |

**Methods for Containment and Clean Up** Keep in suitable, closed containers for disposal. Soak up with inert absorbent material.

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                    |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not ingest. If swallowed then seek immediate medical assistance.                        |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Keep refrigerated. To maintain product quality: Store under an inert atmosphere. Incompatible Materials. Acids. Bases. Water. Strong oxidizing agents. Alcohols. Amines. |

## 8. Exposure controls / personal protection

|                                      |                                                                                                                                                                                                                                                   |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Guidelines</b>           | This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.                                                                                                     |
| <b>Engineering Measures</b>          | Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.                                                                                                 |
| <b>Personal Protective Equipment</b> |                                                                                                                                                                                                                                                   |
| <b>Eye/face Protection</b>           | Tight sealing safety goggles. Face protection shield.                                                                                                                                                                                             |
| <b>Skin and body protection</b>      | Wear appropriate protective gloves and clothing to prevent skin exposure.                                                                                                                                                                         |
| <b>Respiratory Protection</b>        | Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. |
| <b>Recommended Filter type:</b>      | Organic gases and vapours filter. Type A. Brown. conforming to EN14387.                                                                                                                                                                           |
| <b>Hygiene Measures</b>              | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                            |

## 9. Physical and chemical properties

|                                               |                                  |
|-----------------------------------------------|----------------------------------|
| <b>Physical State</b>                         | Liquid                           |
| <b>Appearance</b>                             | Colorless                        |
| <b>Odor</b>                                   | No information available         |
| <b>Odor Threshold</b>                         | No information available         |
| <b>pH</b>                                     | No information available         |
| <b>Melting Point/Range</b>                    | -57 °C / -70.6 °F                |
| <b>Boiling Point/Range</b>                    | 128 °C / 262.4 °F @ 760 mmHg     |
| <b>Flash Point</b>                            | No information available         |
| <b>Evaporation Rate</b>                       | No information available         |
| <b>Flammability (solid,gas)</b>               | Not applicable                   |
| <b>Flammability or explosive limits</b>       |                                  |
| Upper                                         | No data available                |
| Lower                                         | No data available                |
| <b>Vapor Pressure</b>                         | No information available         |
| <b>Vapor Density</b>                          | No information available         |
| <b>Specific Gravity</b>                       | 1.640                            |
| <b>Solubility</b>                             | Decomposes in contact with water |
| <b>Partition coefficient; n-octanol/water</b> | No data available                |
| <b>Autoignition Temperature</b>               | No information available         |
| <b>Decomposition Temperature</b>              | No information available         |
| <b>Viscosity</b>                              | No information available         |
| <b>Molecular Formula</b>                      | C2 Cl4 O2                        |
| <b>Molecular Weight</b>                       | 197.83                           |

## 10. Stability and reactivity

|                               |                                                                     |
|-------------------------------|---------------------------------------------------------------------|
| <b>Reactive Hazard</b>        | None known, based on information available                          |
| <b>Stability</b>              | Stable under normal conditions. Moisture sensitive.                 |
| <b>Conditions to Avoid</b>    | Excess heat. Incompatible products. Exposure to moist air or water. |
| <b>Incompatible Materials</b> | Acids, Bases, Water, Strong oxidizing agents, Alcohols, Amines      |

**Hazardous Decomposition Products** Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Chlorine, Thermal decomposition can lead to release of irritating gases and vapors, Phosgene, Hydrogen chloride gas

**Hazardous Polymerization** Hazardous polymerization does not occur.

**Hazardous Reactions** None under normal processing.

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component                                    | LD50 Oral  | LD50 Dermal | LC50 Inhalation                           |
|----------------------------------------------|------------|-------------|-------------------------------------------|
| Carbonochloridic acid, trichloromethyl ester | Not listed | Not listed  | LC50 = 13.9 mg/m <sup>3</sup> ( Rat ) 4 h |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** Causes burns by all exposure routes

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                                    | CAS No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|----------------------------------------------|----------|------------|------------|------------|------------|------------|
| Carbonochloridic acid, trichloromethyl ester | 503-38-8 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Reacts with water so no ecotoxicity data for the substance is available.

**Persistence and Degradability** Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment.

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

UN-No UN3390  
 Proper Shipping Name TOXIC BY INHALATION LIQUID, CORROSIVE, N.O.S.  
 Hazard Class 6.1  
 Subsidiary Hazard Class 8  
 Packing Group I

#### TDG

UN-No UN3389  
 Proper Shipping Name TOXIC BY INHALATION LIQUID, CORROSIVE, N.O.S.  
 Hazard Class 6.1  
 Subsidiary Hazard Class 8  
 Packing Group I

#### IATA

UN-No UN3390  
 Proper Shipping Name TOXIC BY INHALATION LIQUID, CORROSIVE, N.O.S., FORBIDDEN FOR IATA TRANSPORT  
 Hazard Class 6.1  
 Subsidiary Hazard Class 8  
 Packing Group I

#### IMDG/IMO

UN-No UN3390  
 Proper Shipping Name TOXIC BY INHALATION LIQUID, CORROSIVE, N.O.S.  
 Hazard Class 6.1  
 Subsidiary Hazard Class 8  
 Packing Group I

### 15. Regulatory information

#### United States of America Inventory

| Component                                    | CAS No   | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|----------------------------------------------|----------|------|-----------------------------------------------|-----------------------------|
| Carbonochloridic acid, trichloromethyl ester | 503-38-8 | X    | ACTIVE                                        | -                           |

#### Legend:

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

- - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export

Not applicable

#### International Inventories

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                                    | CAS No   | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL |
|----------------------------------------------|----------|-----|------|-----------|-------|------|------|------|-------|------|
| Carbonochloridic acid, trichloromethyl ester | 503-38-8 | -   | X    | 207-965-9 | X     | X    | X    | X    | X     | -    |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### U.S. Federal Regulations

**SARA 313** Not applicable

**SARA 311/312 Hazard Categories** See section 2 for more information

**CWA (Clean Water Act)** Not applicable

**Clean Air Act** Not applicable

**OSHA - Occupational Safety and Health Administration** Not applicable

**CERCLA** Not applicable

**California Proposition 65** This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations** Not applicable

### **U.S. Department of Transportation**

Reportable Quantity (RQ): N

DOT Marine Pollutant N

DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security** This product does not contain any DHS chemicals.

### Other International Regulations

**Mexico - Grade** No information available

**Authorisation/Restrictions according to EU REACH** Not applicable

| Component                                    | CAS No   | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|----------------------------------------------|----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Carbonochloridic acid, trichloromethyl ester | 503-38-8 | -                                                                   | -                                                                             | -                                                                                                     |

### **Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                                    | CAS No   | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|----------------------------------------------|----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Carbonochloridic acid, trichloromethyl ester | 503-38-8 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                                    | CAS No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|----------------------------------------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Carbonochloridic acid, trichloromethyl ester | 503-38-8 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

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**16. Other information**

|                         |                                                                                                                                                                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prepared By</b>      | Regulatory Affairs<br>Thermo Fisher Scientific<br>Email: EMSDS.RA@thermofisher.com                                                                                                                                                      |
| <b>Creation Date</b>    | 06-Feb-2012                                                                                                                                                                                                                             |
| <b>Revision Date</b>    | 06-Jan-2023                                                                                                                                                                                                                             |
| <b>Print Date</b>       | 06-Jan-2023                                                                                                                                                                                                                             |
| <b>Revision Summary</b> | This document has been updated to comply with the US OSHA HazCom 2012 Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS). |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**